

Mechanistic Interpretability of Vision Transformers: Sparse Autoencoder Feature Decomposition for Visual Concept Discovery

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ABSTRACT

We train Sparse Autoencoders (SAEs) on the MLP output activations of a pre-trained ViT-B/16 at two network depths, and use CLIP as an external cross-modal evaluator to assign interpretable labels to discovered sparse directions—a strategy designed to address the labelling gap that arises in purely visual transformers trained without text supervision. Layer 6 features encode low-level texture and geometric patterns with negligible causal impact (mean relative logit drop: 0.03%); Layer 11 features encompass more semantic concepts and are substantially more causal (mean: 1.40%), concentrated on a single feature (*honeycomb pattern*, 13.49% logit drop under ablation). These results confirm that SAE-based decomposition transfers to purely visual transformers and that external VLM evaluation is a viable labelling strategy.

KEYWORDS

Mechanistic Interpretability, Vision Transformers, Sparse Autoencoders, Polysemanticity, Feature Decomposition, Explainable AI

1 INTRODUCTION

As neural networks are deployed in critical settings, mechanistic interpretability aims to reverse-engineer their computations into analyzable units [6]. Sparse Autoencoders (SAEs) have proven effective at decomposing polysemantic MLP activations of language models into monosemantic features [1, 4]. Applying this to Vision Transformers (ViTs) [5] introduces two specific challenges absent in the language setting: the spatial structure of patch-based attention makes causal tracing harder to formalise, and—crucially—ViTs trained without language supervision provide no built-in mechanism for translating sparse visual directions into human-readable concept labels. This project addresses both challenges by training SAEs on a ViT-B/16 and using CLIP [9] as an external evaluator for cross-modal concept labelling, providing systematic empirical evidence that mechanistic interpretability tools transfer to purely visual transformers.

2 RELATED WORK

2.1 Mechanistic Interpretability in Transformers

Elhage et al. [6] formalised the residual stream framework: the residual stream acts as a central communication channel, with attention heads routing information between tokens and MLP blocks processing within-token representations. Understanding this residual communication is crucial for identifying which layers and components drive specific model behaviours. Building on this, Conmy et al.

[3] introduced ACDC (Automated Circuit Discovery), which identifies the minimal computational subgraph responsible for a target behaviour via iterative activation patching—establishing a rigorous methodology for causal analysis that is, in principle, architecture-agnostic.

2.2 Sparse Dictionary Learning and Monosemanticity

A significant obstacle in mechanistic interpretability is *polysemanticity*—individual neurons activating in response to multiple, unrelated features—attributed to *superposition*, where models encode more features than they have dimensions [6]. SAEs applied to MLP activations or residual stream vectors learn a larger dictionary of sparse, monosemantic features that influence model predictions in predictable ways [1, 4]. These contributions target language models exclusively: tokens are words, features correspond to linguistic concepts, and verifying interpretability amounts to reading the activating tokens.

2.3 Internal Representations of Vision Transformers

Raghu et al. [10] showed that ViTs aggregate global information early via self-attention and propagate features strongly through residual connections—properties with no direct analogue in language models that complicate circuit-level analysis. DINO [2] trains the same ViT architecture through self-distillation, without any label supervision. The resulting representations develop qualitatively different emergent properties—most notably, attention maps that spontaneously segment salient objects—indicating that the training regime shapes internal feature structure substantially. A SAE applied to DINO’s MLP activations would therefore likely yield a different vocabulary of sparse features than the texture- and concept-oriented dictionary we observe in the supervised setting. Our codebase already supports DINOv2 as an alternative backbone, but a systematic comparison across training regimes remains a direction for future work.

2.4 Concept Discovery in Vision-Language Models

Gandelsman et al. [7] decompose CLIP image representations as sparse sums of text-direction embeddings, yielding human-interpretable visual decompositions. Haque et al. [8] apply SAEs to medical VLMs, anchoring discovered concepts to clinical ontologies and evaluating them with an LLM as external judge. Both approaches rest on

pre-existing text-image alignment: the visual concepts are interpretable precisely because they live in a space where text directions are meaningful. In a ViT trained without language supervision, no such space exists—and it is exactly this challenge that this project addresses.

3 RESEARCH GAPS

Three gaps motivate this work. **Systematic SAE analysis for pure vision models:** SAEs have not been applied to ViTs trained without language supervision; visual tokens encode image patches rather than words, and there is no direct way to extract a human-understandable label from a learned sparse direction—making it an open empirical question whether SAEs recover meaningful structure from visual activations. **Spatial structure of ViT attention:** unlike language transformers, ViTs operate over all patch pairs simultaneously with no preferred direction of information flow [10]; tracing how evidence from spatial regions is compressed into the [CLS] token requires causal tools not yet validated in the visual setting. **Cross-modal labelling:** when no internal text-image alignment exists, assigning interpretable labels to SAE features requires an external grounding mechanism—a bottleneck unacknowledged in most vision interpretability work. These gaps converge on one open question: can SAEs recover interpretable, monosemantic features from a purely visual ViT, and how can those features be evaluated without built-in language supervision? This project directly targets that question by using CLIP [9] as a detached external evaluator, decoupling the interpretability analysis from the backbone under study and avoiding any need for a vision-language architecture.

4 METHODOLOGY

Our pipeline has four stages.

Backbone and data. We analyse ViT-B/16 [5] (12 transformer blocks, $d=768$, trained on ImageNet-1k) using 1,000 validation images paired with an 18-concept CLIP vocabulary spanning object categories (*animal, bird, plant, ...*), textures (*fur texture, feather texture, ...*), and geometric patterns (*spotted, honeycomb, scale, ...*).

Activation extraction. Forward hooks collect MLP output activations at Layer 6 (mid-network) and Layer 11 (late-network). Each image yields a 196×768 patch-token matrix per layer ([CLS] excluded), centred via a learned decoder bias following [1].

Sparse Autoencoder. Given activation $\mathbf{x} \in \mathbb{R}^d$, the SAE produces a sparse code

$$\mathbf{z} = \text{ReLU}(\mathbf{W}_{\text{enc}}\mathbf{x} + \mathbf{b}_{\text{enc}}), \quad \mathbf{z} \in \mathbb{R}^m, \quad (1)$$

and reconstructs $\hat{\mathbf{x}} = \mathbf{W}_{\text{dec}}\mathbf{z}$ (unit-norm columns, $m=8d=6144$). Training minimises

$$\mathcal{L} = \|\mathbf{x} - \hat{\mathbf{x}}\|_2^2 + \lambda \|\mathbf{z}\|_1 \quad (2)$$

(Adam, $\text{lr}=10^{-3}$, $\lambda=10^{-3}$, batch 256; 5 epochs for Layer 6, 15 for Layer 11).

CLIP labelling. The $K=5$ highest-activating crops per feature are matched to the vocabulary via mean cosine similarity to “a photo of {concept}” prompts [9]. A feature is characterised as monosemantic

when the CLIP-assigned label is confirmed by visual inspection of the top-activating crops.

Causal verification. Feature k is ablated by subtracting its contribution from patch-token activations: $\mathbf{x}' = \mathbf{x} - f_k(\mathbf{x}) \mathbf{W}_{\text{dec}}[:, k]$. Causal impact is quantified via the Relative Logit Drop (RLD):

$$\text{RLD} = \frac{L_{\text{baseline}} - L_{\text{ablated}}}{|L_{\text{baseline}}|}. \quad (3)$$

A dose-response sweep over ablation strength $s \in [0, 1]$ and a steering experiment ($s=5$, amplifying the feature’s contribution) together rule out artefactual interpretations of the causal signal. All causal metrics are evaluated on a single representative validation image; cross-image robustness is left as future work.

5 EXPERIMENTS AND ANALYSIS

Table 2 reports SAE quality and mean causal impact per layer.

Layer 6 achieves near-perfect reconstruction (3.3% activation density); its mean logit drop of 0.03% confirms that mid-network features are individually causally negligible. Layer 11 has sparser activations (1.8% density) and a mean logit drop 44× higher, concentrated on a single feature. Table ?? lists the ten highest-activating features per layer—a representative sample from the 6,144-entry dictionary, not an exhaustive coverage. Training convergence details for both SAEs are shown in Figure 3.

5.1 Feature Variation Across Depths

Layer 6 features are dominated by low-level texture and geometric patterns (*spotted pattern*: 4 features; *scale pattern*: 2; plus *honeycomb pattern, fur texture, furniture, pineapple*), consistent with the role of mid-network MLP layers as local structure detectors [10]. Visual exemplars for the top-5 Layer 6 features are shown in Figure 2. At Layer 11, the vocabulary shifts towards semantically abstract concepts (*animal*: 2 features; *plant*: 3; *bird*), alongside texture descriptors. Layer 11 also shows markedly sparser activations: $\bar{L}_0 = 107.90$ versus 201.72 at Layer 6, meaning each token is explained by 46% fewer simultaneously active features on average—consistent with increasing specialisation towards the output. Visual exemplars for the top-2 Layer 11 features are shown in Figure 1.

5.2 Causal Analysis

Feature 5065 at Layer 11 (*honeycomb pattern*) accounts for 13.49% relative logit drop under ablation and 5.68% logit increase under amplification; all remaining nine Layer 11 features show drops below 0.4%. This roughly 37:1 ratio (Feature 5065 at 13.49% versus the next highest Layer 11 drop of 0.36% for Feature 5517) indicates a strong concentration of causal weight on a single specialised channel. The spatial heatmap and dose-response curve for Feature 5065 are shown in Figure 4: the heatmap confirms selective activation on coherent image regions, and the monotonically increasing dose-response curve rules out artefactual interpretation.

5.3 Human Evaluation

To validate CLIP-assigned labels beyond automatic scoring, we conducted a small-scale blind evaluation. A human annotator inspected the top-5 activating image crops for each of the ten visualised features without prior knowledge of the CLIP label, and described in

free text the visual pattern shared across the crops. Labels were then compared post-hoc against the CLIP output.

Layer 6. Four out of five features showed at least partial agreement with the CLIP label. Feature 4770 (*spotted pattern*) and Feature 4271 (*honeycomb pattern*) produced the clearest matches: the annotator independently described “small round dots scattered across surfaces” and “repeating geometric cell-like patterns”, respectively. Feature 4506 (*scale pattern*) and Feature 2719 (*spotted pattern*) yielded weaker agreement—the annotator described overlapping ridged surfaces and irregular reddish-brown blotches, which are semantically adjacent to but not identical with the CLIP labels. Feature 2045 (*pineapple*) was the most ambiguous: the annotator noted bumpy or grid-like surface textures and identified pineapple crops explicitly, but the remaining crops (building facade, telescope barrel) did not share a single coherent concept, suggesting the feature may encode a broader structural motif rather than the object class itself.

Layer 11. Agreement was stronger and more consistent. Feature 5065 (*honeycomb pattern*) produced perfect convergence: the annotator described “a hexagonal grid, like a beehive” from crops showing bee honeycombs, a wasp nest, large hexagonal concrete tiles, and a macro plant cell structure—all without prompting. The spatial heatmaps further confirmed that activation was concentrated precisely on the hexagonal regions rather than surrounding context. Feature 3932 (*fur texture*) also matched well; the annotator described “long white rope-like fur” independently identifying the Komondor breed present across all five crops. Feature 1767 (*animal*) and Feature 965 (*plant*) showed label-level agreement but with notable granularity gaps: the annotator described the former as “dark-furred dogs” and the latter as “pineapple texture and spiky leaves”, suggesting that the CLIP labels are underspecific relative to what the features actually encode. Feature 540 (*striped pattern*) was partially matched; the annotator described parallel diagonal lines from staircase railings and banisters, which share the linear regularity of a striped pattern without being stripes in a strict sense.

Overall, 8 out of 10 features received descriptions that were consistent with or more specific than the CLIP label. The two cases of weakest agreement (Feature 4506 and Feature 2045 at Layer 6) both carry the lowest CLIP confidence scores in their layer, supporting the use of confidence as a rough reliability signal. The convergence on Feature 5065—from automatic labelling, human blind review, spatial heatmaps, and causal ablation—provides the strongest evidence of a genuinely monosemantic, interpretable feature among the ten visualised features.

5.4 Discussion

The results address both research gaps. SAEs recover structured, labellable features from purely visual MLP activations: all features at both depths receive distinct CLIP labels, confirmed by visual inspection of top-activating crops. Human assessment independently validated CLIP labels for texture-dominant features; convergence on Feature 5065—from both CLIP scoring and human review, combined with the 13.49% logit drop—constitutes the strongest evidence of monosemantic, causally relevant behaviour. The depth-wise shift

from texture detectors (Layer 6) to semantic features (Layer 11) reproduces the representational hierarchy expected from prior ViT analyses [10]. The concentration of causal impact on a single late-network feature may reflect specialisation in the [CLS] accumulation mechanism, though a larger image sample is needed to confirm whether this pattern is input-specific or systematic. Three Layer 6 features exhibit negative RLD values (Feature 1632: -1.21% , Feature 5536: -0.10% , Feature 4271: -0.04%), indicating that ablating them increases the target logit. This suggests that these features act as inhibitory channels in the mid-network; interpreting their causal role requires analysis beyond the scope of this work.

6 CONCLUSIONS

SAE-based feature decomposition transfers to purely visual Vision Transformers: SAEs trained on ViT-B/16 MLP activations recover sparse, interpretable features at two depths, and CLIP-based external evaluation provides a viable labelling strategy without internal text supervision. Mid-network features (Layer 6) are dominated by texture detectors with negligible causal footprint; late-network features (Layer 11) encompass more semantic concepts, with causal impact concentrated on a single highly specialised feature (Feature 5065, 13.49% logit drop). Limitations include the finite and ImageNet-biased CLIP vocabulary, the restriction to MLP outputs (attention heads and residual stream remain unexplored), the qualitative nature of the monosemanticity assessment, and the restriction of causal evaluation to a single image (RLD and steering metrics may not generalise across the validation set). Natural extensions include circuit-level activation patching to trace classification-specific computation, and a comparative analysis of SAE dictionaries from supervised versus DINO-pretrained ViTs [2]—two regimes known to produce qualitatively different internal representations [10].

Table 2: SAE metrics and mean causal impact. $\bar{L}_0$: avg. active features per patch token.

Layer	R^2	$\bar{L}_0$	Mean RLD
6 (mid)	0.9957	201.72	0.03%
11 (late)	0.9477	107.90	1.40%

Figure 1: Top-2 Layer 11 features (by CLIP confidence): *honeycomb pattern* (5065) and *fur texture* (3932). Each block: contextual crops, full images with highlighted patch, activation heatmaps. Feature 5065 exhibits the highest causal impact (13.49% RLD).

Figure 2: Top-5 Layer 6 features (by CLIP confidence): *spotted pattern* (4770), *honeycomb pattern* (4271), *scale pattern* (4506), *spotted pattern* (2719), and *pineapple* (2045). Each block: contextual crops, full images with highlighted patch, activation heatmaps. These mid-network features encode low-level texture and geometric patterns.

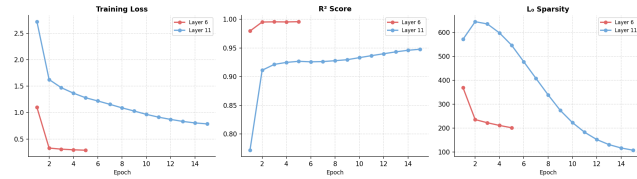


Figure 3: SAE training convergence for Layer 6 and Layer 11. Curves show loss, reconstruction quality (R^2), and sparsity (L_0) over training steps. Both layers converge stably; Layer 6 reaches $R^2=0.9957$ and Layer 11 reaches $R^2=0.9477$, confirming high-fidelity activation reconstruction.

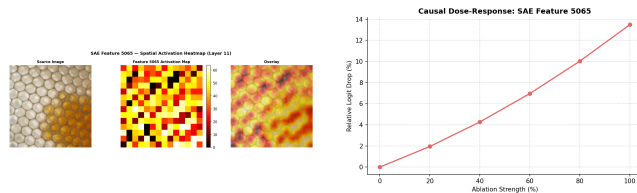


Figure 4: Feature 5065 (Layer 11). Left: spatial activation heatmap showing selective activation on coherent image regions. Right: dose-response curve—monotonically increasing logit drop as ablation strength increases from 0 to 1.

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